Pathway

Take-Home Exercise

Restaurants regularly run an **RFP (Request for Proposal)** process to find the best prices on ingredients from local distributors. Your task is to **build a single, automated system** that runs this entire workflow end-to-end — from parsing a restaurant menu all the way through to emailing distributors for quotes. **Every step below should be executed by your system programmatically, not performed manually.**

Logistics

- **Time limit:** 24 hours from when you receive this exercise.
- **Tech stack:** Use any language, framework, or tools you prefer.
- **Deliverable:** A GitHub repo (or zip) containing your code, a README with setup instructions, and a short Loom / screen recording walking through your solution.
- **Restaurant menu:** Choose any real restaurant menu you like. Include the source (URL or photo) in your submission.
- **UI:** Build a lightweight UI that visualizes each step of the pipeline and its output.

The Exercise

Step 1 Menu → Recipes & Ingredients **Must-have** Given a restaurant menu as input, the system should automatically parse each dish into a structured recipe with ingredients and estimated quantities.

Design a **database schema** to store recipes and their ingredients. All data must be persisted to an actual database (e.g., PostgreSQL, SQLite) — not flat files.

Step 2 Ingredient Pricing Trends (USDA API) **Must-have** The system should automatically query the USDA **API** to fetch recent pricing data for the extracted ingredients.

Store the fetched pricing data in your database alongside the ingredient records. The system should surface pricing trends to help the restaurant understand market context before reaching out to distributors.

Step 3 Find Local Distributors Must-have The system should automatically find **distributors in the restaurant's area** that supply the required ingredients.

Store the identified distributors and their details in your database.

Step 4 Send RFP Emails to Distributors Must-have The system should automatically compose and send **RFP emails** to the distributors found in Step 3, requesting price quotes for the required ingredients.

Each email should include the ingredient(s) needed, estimated quantities, and a deadline for the quote.

The system must be capable of actually sending emails. For the purposes of the demo, you can use mock email addresses instead of real distributor addresses.

Step 5 Collect & Compare Quotes Nice-to-have The system should include an **agent that monitors an inbox** for replies from distributors, automatically parses the quoted prices, and compiles a comparison table.

The agent should handle back-and-forth autonomously — for example, sending follow-up emails if a quote is missing information.

The system should produce a final recommendation of which distributor to go with, based on price, delivery terms, or other factors.

What We're Looking For

- **System design** — How you architect the pipeline, design the database schema, and connect the steps together.
- **Code quality** — Clean, readable, well-structured code with sensible abstractions.
- **Problem-solving** — How you handle ambiguity, edge cases, and real-world messiness (e.g., menus with vague dish names, missing distributor data).

> There are no trick questions here. We care more about how you think and build than whether every API call works perfectly. If you hit a wall with a particular API or service, document your approach and move on. Good luck — we're excited to see what you build!